Nishit Rana

Founding Engineer · AI Systems, Backend & Infrastructure

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Summary

Builder who ships the whole stack alone: LLM agents and eval harnesses, the Postgres schemas and job queues behind them, the payment rails that monetise them, and the AWS servers they run on. Graduated in 2025; within six months went from intern to **founding engineer at Cognerd**, owning product engineering end to end—**20+ APIs, three durable job systems, Razorpay billing, grounded agents**, and a **migration of four production services onto self-managed EC2/RDS** defined as code. Lead technical delivery and supervise three interns. Ship open-source agent tooling on the side.

Experience

Founding Engineer (SDE 1) | Cognerd *Feb 2026 – Present*

- **Ownership.** Own planning, build, release, deployment and production support across all company platforms; lead technical delivery and supervise three interns with end-to-end responsibility.

AI & AGENT SYSTEMS

- **Grounded agents.** Built tool-using PydanticAI agents with retrieval budgets, parallel-call locking, model timeouts, citations, spend admission and per-caller usage accounting; added sentence-level claim/citation verification with controlled fallbacks for multilingual replies.
- **Evaluation.** Built golden datasets, labelling/export tooling and baseline regression gates for brand matching, sentiment, competitor detection and share-of-voice / visibility scoring.
- **LLM pipelines.** DataForSEO-grounded prompt generation (search intent, People-Also-Ask, personas, regional/language targeting); prompt library with versions, SSE streaming, provider-aware retries and interrupted-run recovery; encrypted bring-your-own-key provider selection.
- **Generative video.** n8n + Vertex AI Veo product-to-video pipeline: Gemini image analysis, structured prompts, async polling, safety-block handling, R2 upload and backend callbacks; productised as a UGC video studio with recovery, history and billing controls.
- **MCP & human-in-the-loop.** Dashboard-based MCP consent, browser-bound account linking and metered usage; reviewed drafts, human-takeover views, per-block source lineage and evidence boundaries on personalisation.

BACKEND, DATA & PAYMENTS

- **APIs & modelling.** Designed 20+ REST routes and 10+ PostgreSQL tables for billing, analysis, publishing, notifications, agents and source provenance.
- **Durable job systems.** Built three on PostgreSQL + BullMQ (SEO audits, weekly brand analysis, scheduled social publishing) with atomic admission, delivery ledgers, account leases and crash recovery.
- **Database performance & correctness.** Replaced unbounded reads with cursor pagination; added compound indexes, advisory-locked transactions, idempotency keys, expiring nonces and uniqueness constraints.
- **Redis / Valkey.** Shared counters, rate limits and generation-fenced caching with admission limits and degradation handling; replaced full cache scans on every write with bounded checks.
- **Payments.** Razorpay checkout, signature verification, webhook de-duplication and reconciliation; migrated legacy credits to subscription allowances, feature gates, refunds and overrides.
- **Auth & security.** Better Auth flows, multi-tenant Viewer/Editor/Admin RBAC, fixed cross-user data-access paths; Zod validation, rate limits, SSRF-protected audit runner, sanitised errors, request IDs.

INFRASTRUCTURE & SERVER MANAGEMENT

- **AWS migration.** Moved four production services and PostgreSQL to EC2/RDS behind ALB; verified restores, DB roles and TLS; run Linux/systemd services, logs and incident recovery.
- **Infrastructure as code.** CloudFormation for CodePipeline/CodeBuild/CodeDeploy with scoped IAM; automated host preparation, packaging, runtime assets and systemd units.
- **Release safety.** Migration preflights, service-user execution, health validation, replay draining, signal-safe hooks; secrets via Secrets Manager and SSM.
- **Networking & observability.** HTTPS routing and access boundaries with Cloudflare, ALB and ACM; CloudWatch on EC2/ALB/RDS with SNS alerts; ElastiCache Valkey with TLS, ACLs, alarms and recovery drills.
- **Cost.** S3/CloudFront delivery with origin access control and Vercel OIDC; cut dev-infra spend via Nginx consolidation, office-hours EC2 scheduling and removal of redundant network resources.

PRODUCT & INTEGRATIONS

- **Growth stack.** Composio-based LinkedIn/X publishing and Google Calendar sync; GA4, Search Console and Clarity dashboards; Google Ads campaign management with guarded, partial-failure-safe actions.
- **Multi-tenant CMS.** Built CogCMS (tenant roles, scoped API keys, CSRF, webhooks) on MongoDB with dry-run importers and parity checks; migrated the marketing site to Next.js.

Backend Developer Intern | Welzin *Aug 2025 – Jan 2026*

- Built Node.js/Python APIs for brand monitoring: user-scoped brand profiles, persona-based generation, AEO reports and AI-provider comparison; integrated OpenRouter/OpenAI/Grok with MongoDB, RAG and n8n.
- Deployed Linux applications behind Nginx; worked with Docker/Kubernetes; moved a Flask service to Vercel.

Open-Source & Projects

AgentGraph — Agent Observability & Tool Intelligence https://github.com/nrana-pixel/AgentGraph

TypeScript · Python · Next.js · PostgreSQL · Redis · OpenTelemetry

- Trace/span ingestion, waterfall timelines and token/cost analytics; TypeScript and Python SDKs with OpenAI/Anthropic instrumentation, async flushing, bounded retries and redaction.
- Trace-driven tool-swap recommendations, monthly spend caps, signed cost alerts, OTLP and S3 NDJSON export.

AGXP — Cloudflare Edge Delivery for AI Crawlers https://github.com/nrana-pixel/agxp-cloudflare

Built at Cognerd · TypeScript · Hono · Workers · D1 · KV · React

- Proposed and built a platform serving LLM-transformed HTML variants to AI crawlers from Workers/KV with origin fallback; automated Worker deployment, KV provisioning and routes; AES-256-GCM token encryption.

Optimus CLI — Terminal AI Coding Assistant https://github.com/nrana-pixel/optimus-cli

TypeScript · oclif · React/Ink · MCP · Ollama

- MCP client (stdio, tool discovery, invocation), repository indexing, Ollama/OpenAI-compatible streaming models.

Jarvis — Personal AI Workflow OS https://github.com/nrana-pixel

Python · PowerShell · Bash · Claude Code · Codex

- Git-backed agent memory with provenance-aware admission; isolated writer/reviewer agents, regression runners and baseline drift checks.

Also: Raja Bole (deterministic TypeScript game engine with bot simulation harness); interactive Next.js/Web Audio portfolio with FFT visualisation.

Technical Skills

AI Engineering	PydanticAI, MCP, RAG, Vertex AI / Veo, Gemini, OpenAI, Anthropic, structured outputs, grounding, eval datasets & regression gates
Backend	TypeScript, Python, Go, SQL, Node.js, Express, Hono, FastAPI, Next.js, React, REST, SSE
Data & Queues	PostgreSQL, MongoDB/Atlas, MySQL, Cloudflare D1, Qdrant, Redis/Valkey, BullMQ, n8n
Infra & Ops	AWS (EC2, RDS, ALB, ACM, S3, CloudFront, ElastiCache, IAM, Secrets Manager, SSM, CloudWatch, SNS, CloudFormation, EventBridge, CodePipeline), Cloudflare Workers/KV/R2, Linux, systemd, Nginx, Docker, Kubernetes, Tailscale, Vercel
Testing & Tools	Vitest, Jest, pytest, Playwright, Git, Bash, PowerShell, Claude Code, Codex

Education & Community

Bachelor of Computer Applications | Arya College, Ludhiana *2022 – 2025*

- **Speaker**, "kagent for beginners", Cloud Native Community Group (Jul 2026) — live demo and limitations.
- **First Prize**, Swift Surfer, Technothan 2022 — PG Dept. of CS, Guru Nanak Khalsa College.